Assignment # 1

Analyzing and Troubleshooting a Home Wi-Fi Network

Due Date: October 31, 2025

Task 1: Network Mapping

Connect **2–3 computers** (and optionally a smartphone) to the same Wi-Fi network.

On each computer, run the following commands:

```
ipconfig
ping <gateway IP>
arp -a
tracert 8.8.8.8
```

Note:

- IPv4 address
- Default gateway
- Subnet mask
- MAC address
- Traceroute result (number of hops to Google DNS)

Task 2: Network Scanning

Install and use a **network scanner tool** (choose one):

- **Angry IP Scanner** (Windows/macOS/Linux)
- **Advanced IP Scanner** (Windows)
- **Fing** (Android/iOS)

Scan your local network and list **all connected devices** (with IP, MAC, and device names if available).

Task 3: Wi-Fi Router Exploration

Log in to your router's web interface (use the gateway IP in a browser, e.g., http://192.168.1.1).

Find and note:

- DHCP range (e.g., 192.168.1.2–192.168.1.100)
- Number of connected devices
- Wireless channel and frequency (2.4 GHz or 5 GHz)

Task 4: Connectivity and Troubleshooting

Try disconnecting one computer and reconnecting — does it get a new IP?

Disable Wi-Fi and connect using a mobile hotspot — compare IP addresses.

Identify one possible reason if **ping** fails between two devices (e.g., firewall, subnet mismatch, ICMP blocked).

Deliverables

Prepare a short report (3–4 pages) as a PDF file that includes:

- A **network diagram** showing a router, PCs, and their IPs.
- Screenshots of:
  - `ipconfig` and `ping` outputs
  - Network scan results
  - Router info page
- A short paragraph explaining how DHCP assigns IP addresses.
- Observations and one troubleshooting insight.